PS-31 · SATELLITE INFRASTRUCTURE DAMAGE ANALYZER

SDG 9 · 11 · 13

1 PROBLEM

- Cyclone / flood / quake hits a town
- Manual inspection takes DAYS
- Nobody knows where to go first

Faces it: authorities & rescue teams



2 PROPOSED SOLUTION

- Web app: upload BEFORE + AFTER images
- AI finds damaged buildings & roads
- Ranked inspect-first list + exact GPS

★ UNIQUE: rank = damage × importance × cluster

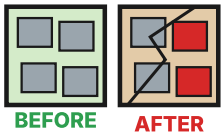
3 SOLUTION FLOW (how it works)

USER



Disaster officer / rescue team

INPUT / DATA

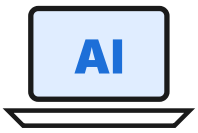


Pair of satellite images

PROCESSING

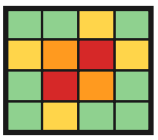
1 Align images
2 Spot changes
3 Rate damage
(computer vision)

CORE SYSTEM



Damage scoring + Priority Index

OUTPUT



Damage map + GPS priority list

4 MODULE STRUCTURE

SATELLITE INFRASTRUCTURE DAMAGE ANALYZER

Data Module
- Upload before / after
- Check image quality
- Align + read GPS tags

images

AI Damage Module
- Spot changed areas
- Score each building
- Flag blocked roads

labels

Priority + GPS Module
- Priority Index ranking
- Exact GPS + route link
- Export PDF / CSV / KML

5 USER / SYSTEM FLOW

1. Open app, upload BEFORE + AFTER
2. System checks & aligns images
3. AI scores damage per building
4. Map + GPS priority list appear
5. Get field route & report

6 TECHNICAL APPROACH

Frontend
Streamlit web app + live map

Backend
Python: OpenCV, scikit-image

AI / CV layer
Align + change score (U-Net optional)

Geo data
GeoTIFF pairs + OSM footprints

Report
CSV · GeoJSON · KML · PDF

7 KEY FEATURES

- AI damage rating per site
- Exact GPS of worst sites
- Critical sites ranked first
- Field route + map link
- Report: PDF / CSV / KML

8 EXPECTED OUTPUT

- Colour damage map red / orange / yellow
- Priority list with GPS (lat, long) + map link
- Field route + report (PDF / CSV / KML)

9 CONSTRAINTS

- Sees visible damage only, not structural safety
- Needs GPS-tagged, same-area image pair
- Cloud / low resolution lowers accuracy
- Public imagery, 20-hr build

PAGE 2 · HOW TO DRAW & PRESENT (private, do not submit)

A. Ruler guide: where each box goes (mm from the top-left corner of A4)

Block	Left	Top	Width × Height
Title bar	7	5	196 × 10
1 Problem | 2 Solution	7 | 107	18	96 × 36 each
3 Solution Flow (5 boxes inside, each 32 wide, arrows 7 apart)	7	57	196 × 49
4 Module Structure (top box 110 wide, 3 modules 55 wide)	7	109	196 × 54
5 User Flow | 6 Technical	7 | 107	166	96 × 76 each (5 boxes, 10 tall, 3 gap)
7 Features | 8 Output | 9 Constraints	7 | 74 | 141	245	62 × 45 each

Leave a 7 mm margin on every side. Every header strip is 7 mm tall. Rounded corners are optional.

B. Pen plan (you need only 1 black pen + 7 colours)

- **Black fineliner:** all text, arrows, icons
- Red: 1 Problem
- Blue: 2 Solution, Data module
- Purple: 3 Flow
- Teal: 4 Modules
- Indigo: 5 User flow
- Brown: 6 Tech
- Gold: 7 Features
- Green: 8 Output, Priority module
- Grey pencil: 9 Constraints
- Orange: AI Damage module. Map squares: red / orange / yellow / green

Colour only the header strips and the outlines. Do not fill whole boxes: it takes too long and smudges.

C. Draw order (about 45 min)

1. Pencil the 5 rows with a ruler (use table A)
2. Pencil header strips + the number circles 1–9
3. Row 3 flow boxes + arrows (the hero: spend time here)
4. Row 4 tree + 3 modules
5. Blocks 5 and 6 (copy the 5 boxes and arrows)
6. Bottom row 7 / 8 / 9
7. Ink all text in **capital block letters**, 1 cm tall for titles
8. Colour, then erase the pencil

Easy icons: person = circle + hat + half-circle; images = 2 squares with small boxes; laptop = rectangle + "AI"; map = 4×4 grid coloured in.

D. Your 60-second walkthrough (point at each number as you speak)

1. ① **Problem:** "After a cyclone or flood, teams inspect roads and buildings by hand. It takes days, and nobody knows where to go first."
2. ② **Solution:** "Our web app takes a BEFORE and AFTER satellite image, finds the damage, and gives a ranked inspect-first list with exact GPS. *Unique:* we rank by damage × how important the place is × how badly the neighbourhood is hit, so a damaged hospital outranks a damaged house."
3. ③ **Flow:** "An officer gives two images; we align them, spot changes, score each building, and output a damage map and GPS priority list."
4. ④ **Modules:** "Data cleans and geo-reads the images, AI Damage scores each building and flags blocked roads, Priority + GPS ranks sites and exports."
5. ⑤ **User flow:** "Five steps from upload to a field route and report."
6. ⑥ **Tech:** "Streamlit app, Python with OpenCV, GeoTIFF imagery, OpenStreetMap footprints; exports to CSV, KML for phones, and PDF."
7. ⑦⑧⑨: "Output: colour map, GPS list, route. Limit: visible damage only, so engineers still verify on site."

E. Questions evaluators may ask

Q: What is unique? Most tools stop at a heat-map. We add exact GPS per site, a Priority Index (damage × facility importance × neighbourhood damage), a field route with a Google Maps link, and an evidence card explaining each rank.

Q: How do you detect damage? We align the pair, then for each pre-event building footprint measure how much of the roof changed, whether structure/edges survived, and blend these into a 0–1 score → none / minor / major / destroyed. A trained U-Net (xBD) is our upgrade path.

Q: Where do officers get images? From ISRO/NRSC, the International Charter, Copernicus EMS rapid mapping, or open programmes like Vantor Open Data. Our input is any GPS-tagged GeoTIFF pair.

Q: How accurate? We measure it (confusion matrix, top-10 precision). On our synthetic test: ~91% exact, 90% top-10 precision. Real-data validation on xBD is the next step.

Q: Roads? We flag road segments with unusual change as "possible obstruction"; they need field verification.

Q: Why not Sentinel? 10 m pixels are too coarse for single buildings; fine for flood extent only. We use 0.3–0.8 m imagery.

Q: Who benefits? Disaster authorities, rescue teams, local government. SDGs 9, 11, 13.